

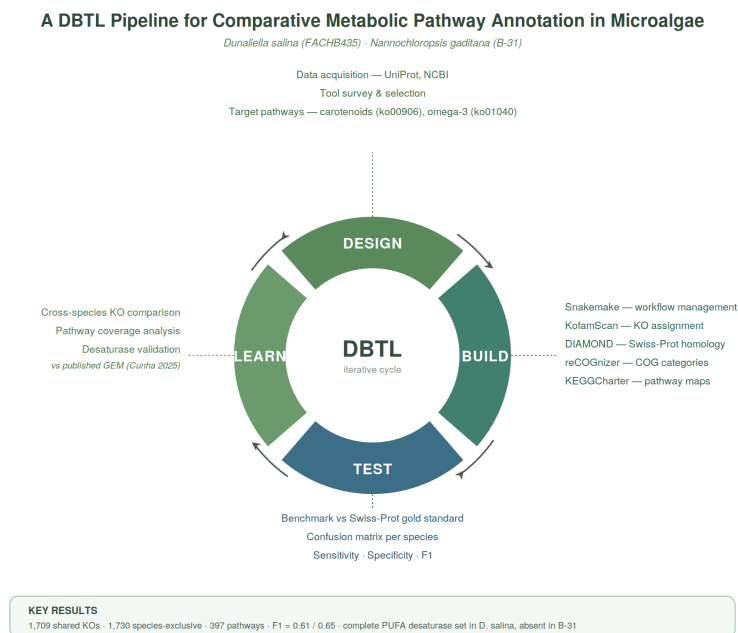
A DBTL-based Bioinformatics Pipeline for Comparative Metabolic Pathway Annotation in Microalgae

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Abstract. Microalgae are a promising yet underexplored resource for the sustainable bioproduction of high-value compounds such as carotenoids and omega-3 fatty acids, but comparative metabolic pathway annotation across species is hampered by data heterogeneity, incomplete genome annotations, and the absence of reproducible analytical frameworks. This work presents an open-source pipeline that applies the Design–Build–Test–Learn (DBTL) paradigm to the comparative annotation of *Dunaliella salina* and *Nannochloropsis gaditana*, and reports the comparative results obtained. The pipeline integrates KofamScan (KEGG Orthology assignment) and DIAMOND (homology annotation against Swiss-Prot) within a Snakemake workflow, complemented by reCOGNizer and KEGG-Chart. Applied to 16,697 and 10,929 proteins respectively, it assigned KEGG Orthology identifiers (KOs) to 20.2% and 29.3% of proteins, evaluated 397 KEGG pathways, and identified 1,709 shared and 1,730 species-exclusive KOs. Benchmarking against a Swiss-Prot gold standard yielded F1 scores of 0.61 and 0.65 with high specificity (0.93 and 0.88), indicating conservative but reliable KO assignment. Targeted validation of fatty-acid desaturases against a published genome-scale metabolic model confirmed that *D. salina* encodes a complete poly-unsaturated fatty-acid desaturase set, whereas the *N. gaditana* B-31 assembly lacks the Δ^4 - and ω^3 -desaturases reported for strain CCMP1894—a strain-level annotation gap rather than a true biological absence. All outputs were deposited in a public repository, directly addressing the reproducibility and cross-species comparability gaps identified in the literature.

Keywords: Microalgae · DBTL Cycle · Metabolic Annotation · Bioinformatics Pipeline · Reproducibility · KEGG Orthology



1 Introduction

Microalgae are unicellular photosynthetic organisms whose metabolic versatility makes them attractive platforms for the sustainable production of high-value compounds [2]. Commercially relevant products include pigments such as β -carotene, poly-unsaturated fatty acids (PUFAs) such as eicosapentaenoic acid (EPA), and bioactive secondary metabolites, all accumulable at industrially viable concentrations [3]. Two species have complementary biotechnological profiles. *Dunaliella salina* is a halotolerant green alga that accumulates carotenoids to more than 10% of its dry weight under salt stress [2, 5]. *Nannochloropsis gaditana* is a marine eustigmatophyte with high EPA content and rapid lipid accumulation, for which a genome-scale metabolic model (GEM) has been re-constructed [4].

Despite these profiles, a rigorous, quantitative comparison of the metabolic pathway repertoires of the two species is missing from the literature. Existing annotations are fragmented across heterogeneous databases, lack nomenclature standardisation, and were produced with different tools and reference sets, making cross-species inference unreliable. The DBTL paradigm, formalised for microbial biosynthesis [1], provides a structure for addressing this: its four phases—iterative computational design, workflow construction, empirical validation, and data-driven refinement—map naturally onto a bioinformatics pipeline. This paper presents such a pipeline and, with emphasis on the comparative findings, reports the results obtained for the two species.

The central problem is that incomplete and heterogeneous annotations prevent reliable cross-species comparison in microalgae, impeding identification of species-specific biosynthetic capabilities. Four SMART objectives were defined: (1) survey and critically evaluate ≥ 6 annotation, pathway-prediction and metabolomics tools, building a comparison matrix (scope, I/O, licence, API, containerisation, interoperability); (2) integrate two complementary tools into a reproducible Snakemake workflow deposited in a public repository; (3) annotate and compare the metabolic repertoires of the two species, focusing on carotenoid and omega-3 biosynthesis; and (4) benchmark accuracy using sensitivity, specificity and F1 against curated Swiss-Prot annotations and published literature.

2 State of the Art

2.1 The DBTL Framework

The DBTL cycle was introduced as an engineering abstraction for optimising microbial biosynthetic pathways [1]: *Design* enumerates candidates *in silico*; *Build* constructs them; *Test* quantifies performance; and *Learn* extracts insight that seeds the next iteration. Its four phases translate cleanly onto a computational annotation workflow, where Build denotes workflow construction and Test denotes benchmarking against curated references.

2.2 Bioinformatics Tools and Prior Work for Microalgae

A growing ecosystem of open-source tools supports functional annotation and metabolic modelling of microalgae. The UPIMAPI, reCOGnizer and KEGGCharacter suite [8] provides homology- and domain-based annotation and KEGG pathway visualisation under reproducible command-line interfaces, and underpins the MOSCA meta-omics pipeline [9]; genome-scale metabolic reconstruction is supported by platforms such as *merlin* [10, 11]. From this landscape, the present work integrates a focused subset into a reproducible annotation pipeline, described below.

The pipeline builds on five established open-source tools:

- **Snakemake** [7] — Python-based workflow management with automatic dependency resolution.
- **KofamScan** [13] — KEGG Orthology (KO) assignment from family-specific HMM profiles with adaptive thresholds.
- **DIAMOND** [12] — fast protein homology search against the Swiss-Prot reference.
- **reCOGnizer** [8] — domain-homology annotation against COG and related databases.
- **KEGGCharacter** [8] — projection of annotated genes onto KEGG metabolic maps.

2.3 Identified Gaps

None of these efforts shares a methodology or benchmarking framework, leaving three gaps that motivate this work: **data heterogeneity** (incompatible databases and identifiers), **lack of comparative reproducibility** (no standardised cross-species workflow), and **insufficient benchmarking** on microalgal genomes, which are rich in hypothetical proteins. To satisfy Objective 1, candidate tools were scored with a structured comparison matrix prior to selection (Section 3); the full per-phase matrices accompany the project repository as supplementary material.

3 Materials and Methods

The pipeline maps explicitly onto the four DBTL phases. All code was written in Python 3.12 (Biopython, pandas, Matplotlib) and is available in a public GitHub repository under an MIT licence. The architecture is deliberately modular: each analytical step is an independent Snakemake rule with declared inputs and outputs, so that any single tool can be replaced or extended without disrupting the remainder of the workflow. Intermediate results—in particular the KEGG REST responses, which dominate wall-clock time—are cached on disk, so that re-execution after a parameter change recomputes only the affected steps. This design directly targets the reproducibility gap identified in Section 2: a third party can clone the repository, install the pinned dependencies, and regenerate every figure and table in this paper from the raw protein FASTA files with a single command.

Data. Protein sequences for *D. salina* (strain FACHB435 [5]) were retrieved from UniProtKB (taxon 3046), as the FACHB435 genome lacks a direct protein-download endpoint; *N. gaditana* proteins were retrieved from NCBI BioProject PRJNA170989 (strain B-31), the only programmatically accessible set (Table 1).

Design. Two complementary annotation strategies were selected so that one tool provides pathway-level orthology while the other provides an independent curated gold-standard: profile-HMM KO assignment (KofamScan) and homology annotation (DIAMOND against Swiss-Prot). reCOGnizer (COG categories) and KEGGCharter (visualisation) complete the stack. Tool selection followed a structured comparison matrix scoring each candidate on functional scope, input/output formats, open-source licensing, API availability, containerisation and interoperability (Objective 1); the complete per-phase matrices are provided as supplementary material in the project repository. Carotenoid (ko00906) and unsaturated fatty-acid (ko01040) biosynthesis were defined as target pathways.

Build. The workflow was implemented as a Snakemake directed acyclic graph with one rule per step, integrating two annotation tools as formal rules (Objective 2). KofamScan [13] scores each protein against family-specific HMM profiles

with adaptive thresholds, reporting only above-threshold (significant) KO assignments; each KO represents a specific biochemical function and links a protein to the pathways in which it participates. DIAMOND [12] `blastp` was run against Swiss-Prot (~570,000 reviewed entries) in sensitive mode ($e \leq 10^{-5}$, best hit by bit score). reCOGnizer assigned COG categories. For each significant KO, the KEGG REST API was queried (with local caching) to retrieve participating pathways, and KEGGCharter rendered the target maps.

Test. Accuracy was benchmarked treating the curated Swiss-Prot result as a gold standard: a protein is a gold-standard positive if it has a Swiss-Prot hit with $\geq 30\%$ identity and $\geq 50\%$ query coverage, and a predicted positive if KofamScan assigns it a significant KO. From the resulting confusion matrix (TP/FP/TN/FN), sensitivity = $TP/(TP+FN)$, specificity = $TN/(TN+FP)$, precision = $TP/(TP+FP)$ and $F1 = 2 \text{Prec} \cdot \text{Sens}/(\text{Prec}+\text{Sens})$ were computed.

Learn. Three analyses extracted comparative insight: *pathway coverage* (fraction of reference KOs present per pathway, classified *complete* $\geq 50\%$, *partial* 10–49%, *absent* $< 10\%$); *cross-species comparison* (intersection of KO sets); and *targeted validation* of the fatty-acid desaturase complement against the *Nanochloropsis* GEM of Cunha [6].

4 Results

4.1 Functional Annotation Yield

KofamScan assigned KOs to a minority of proteins in both species, with a higher assignment rate in *N. gaditana* than in *D. salina*. DIAMOND against Swiss-Prot annotated a larger fraction than KofamScan in both species, again with the higher annotation rate in *N. gaditana*; the median identity and query coverage of the valid hits were comparable between the two species (Table 1). The consistently higher annotation yield for *N. gaditana* across both tools tracks the provenance of its proteome (a curated NCBI BioProject) rather than any biological difference—a point developed in Section 5. reCOGnizer assigned COG categories to 4,877 proteins, the most populated being translation, energy production and amino-acid metabolism.

The most frequent KO functions revealed contrasting proteome composition. The large eukaryotic genome of *D. salina* was dominated by chromatin-packaging histones (H2A, H3, H4), consistent with the larger nuclear genome and more extensive chromatin machinery expected of a green alga, while the compact genome of *N. gaditana* was dominated by metabolic enzymes (3-hydroxybutyrate dehydrogenase, glutathione S-transferase), reflecting a streamlined genome in which a larger proportion of coding capacity is devoted to core metabolism. In both species, “uncharacterised protein” was the single most frequent annotation—a direct measure of the functional dark matter that motivates the work, and a reminder that even well-studied microalgae retain a substantial fraction of genes of unknown function.

Table 1. Input datasets and functional annotation yield. Percentages are relative to each proteome.

	<i>D. salina</i> FACHB435	<i>N. gaditana</i> B-31
Source	UniProtKB (taxon 3046)	NCBI (PRJNA170989)
Proteins	16,697	10,929
KofamScan KOs	3,366 (20.2 %)	3,199 (29.3 %)
DIAMOND annotated	42.2 %	51.8 %
median identity	45.6 %	43.2 %
median query coverage	84.0 %	80.0 %

The discrepancy between the two annotation strategies is itself informative. DIAMOND annotated roughly twice as many proteins as KofamScan in each species, which is expected: homology search against a large curated database will return a hit for many proteins that lack a confident match to a KO-specific HMM profile. However, a DIAMOND hit to a curated protein does not by itself assign a pathway role, whereas a KofamScan KO does. The two tools therefore occupy complementary positions on a precision–recall trade-off (Section 4.6): KofamScan favours precision and pathway-level interpretability, DIAMOND favours recall and breadth of annotation.

4.2 Cross-species Metabolic Repertoire

Intersecting the KO sets identified 1,709 shared KOs, 907 exclusive to *D. salina* and 823 exclusive to *N. gaditana* (Fig. 1). The shared core represents roughly two-thirds of each species’ KO set (65.3 % of the *D. salina* KOs and 67.5 % of the *N. gaditana* KOs) and reflects conserved central metabolism—glycolysis, the citric-acid cycle, nucleotide and amino-acid biosynthesis, and the core photosynthetic apparatus. The exclusive fractions point to species-specific capabilities consistent with their distinct ecological niches: a green alga (Chlorophyta) adapted to hypersaline environments, in which osmotic protection and carotenoid-mediated photoprotection are prominent, versus a marine, oleaginous eustigmatophyte (Ochrophyta) optimised for lipid storage. Because the two species belong to distantly related lineages, this shared-versus-exclusive partition also provides a coarse, annotation-level view of metabolic conservation across a deep phylogenetic split, complementing the gene-level comparisons reported in prior single-species studies [4, 5].

4.3 Pathway Coverage

Across 397 evaluated pathways, *D. salina* presented 11 complete, 166 partial and 220 absent, and *N. gaditana* 8 complete, 178 partial and 211 absent (Fig. 2). Pathways were scored by the fraction of their reference KOs detected and labelled

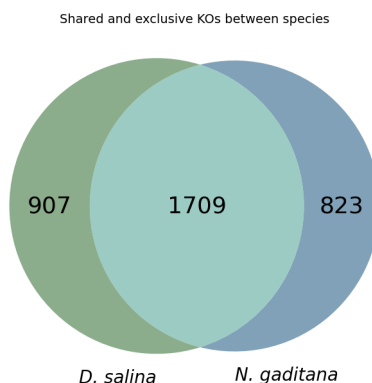


Fig. 1. Overlap of KEGG Orthology assignments between the two species.

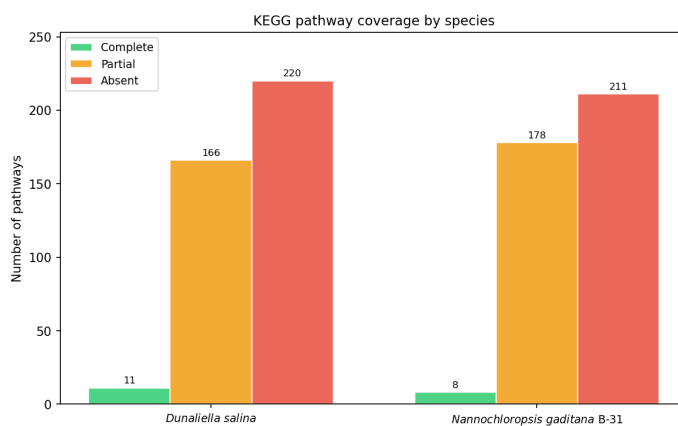


Fig. 2. Pathway coverage distribution by species.

complete ($\geq 50\%$), partial (10–49%) or absent ($< 10\%$). The predominance of partial coverage is expected rather than a sign of missing metabolism: KEGG reference pathways pool orthologues from bacteria, plants, fungi and animals and often bundle alternative routes to the same product, so a single microalga is not expected to carry every reference KO. A 50% presence is therefore a reasonable mark of an operational pathway, while the lowest fraction captures pathways represented by only one or two incidental KOs. Both target pathways were partial in both species—carotenoid biosynthesis at 12/61 KOs and unsaturated fatty-acid biosynthesis at 6/32 KOs in *D. salina*—whereas glycolysis was complete in both, as expected for a conserved central pathway.

Table 2. Fatty-acid desaturases detected by KofamScan, validated against the *Nannochloropsis* GEM of Cunha [6].

KO	Enzyme	<i>D. salina</i>	<i>N. gaditana</i>	Expected (GEM)
K00507	$\Delta 9$ -desaturase	present	present	both
K10255	$\omega 6$ -desaturase	present	absent	<i>D. salina</i>
K10256	$\omega 6$ -desaturase	present	present	both
K10257	$\omega 3$ -desaturase	present	absent	both
K12418	$\Delta 4$ -desaturase	present	absent	<i>N. gaditana</i> CCMP1894
K21735	(8-3)-desaturase	absent	present	<i>N. gaditana</i> B-31 only

4.4 Target Pathways: Carotenoids and Omega-3

Within carotenoid biosynthesis, KofamScan recovered key enzymes in *D. salina* including phytoene desaturase, ζ -carotene desaturase, ζ -carotene isomerase and zeaxanthin epoxidase, consistent with the species’ documented β -carotene over-accumulation. KEGGCharter renderings of both target pathways (per-species “potential” maps and a cross-species “differential” map) are shown in Fig. 3, where colour encodes the number of annotated genes mapped to each pathway step. The visualisation confirms denser coverage of the carotenoid backbone in *D. salina* and highlights the steps differentiating the two species.

4.5 Desaturase Validation against a Published GEM

The fatty-acid desaturase complement, central to omega-3 biosynthesis, was examined in detail and cross-checked against the desaturases reported by Cunha [6] (Table 2). *D. salina* encoded a complete PUFA desaturase set: $\Delta 9$ - (K00507), $\omega 6$ - (K10255, K10256), $\omega 3$ - (K10257) and $\Delta 4$ -desaturases (K12418). By contrast, *N. gaditana* B-31 encoded only the $\Delta 9$ -desaturase (K00507), an $\omega 6$ -desaturase (K10256) and an (8-3)-desaturase (K21735); the $\Delta 4$ - and $\omega 3$ -desaturases were absent. Because these enzymes are confirmed present in CCMP1894 [6], their absence in B-31 is most parsimoniously explained by the known annotation gaps of that assembly rather than a true biological absence—a result with direct implications for strain selection (Section 5). This cross-referencing of an automated annotation against a curated GEM is the clearest demonstration of the Learn phase’s value.

4.6 Benchmarking Accuracy

Benchmarking KofamScan against the Swiss-Prot gold standard (Tables 3 and 4) yielded F1 scores of 0.61 (*D. salina*) and 0.65 (*N. gaditana*). Both species showed high specificity (0.93 and 0.88) but moderate sensitivity (0.53 and 0.61): when KofamScan assigns a KO it is rarely a false positive, but it annotates fewer proteins than the homology search—consistent with its adaptive thresholds, which

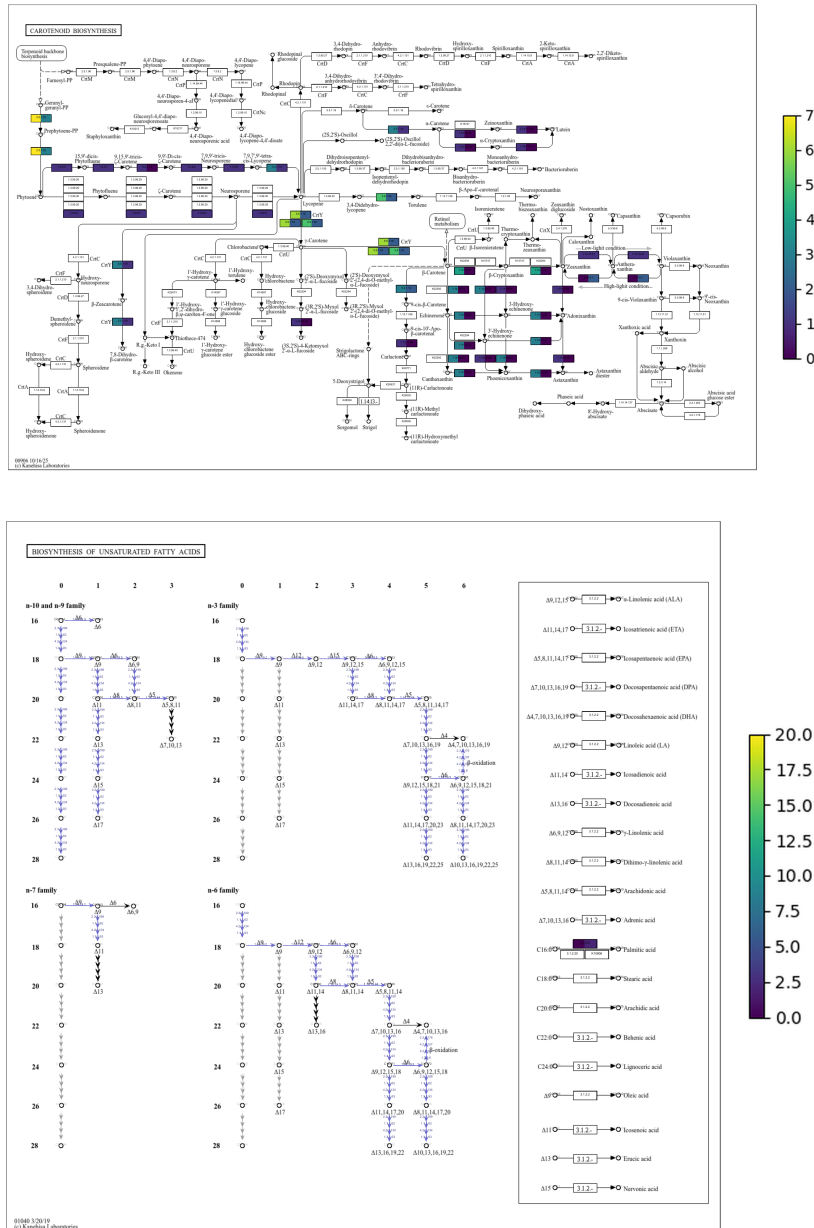


Fig. 3. KEGGCharter differential pathway maps comparing the two species for the two target pathways: carotenoid biosynthesis (ko00906, top) and unsaturated fatty-acid biosynthesis (ko01040, bottom). Colour encodes the number of annotated genes mapped to each pathway step.

are designed to suppress spurious assignments. The two species present complementary profiles: *N. gaditana* achieves higher sensitivity and F1, while *D. salina* achieves higher specificity and precision. Both annotators therefore perform comparably overall, with strengths distributed across different metrics, which argues for retaining both: the high-specificity KO assignments form a confident backbone, while the broader homology search captures proteins lacking a confident HMM match.

Table 3. Confusion matrix for *Dunaliella salina*. KofamScan (predicted) versus the Swiss-Prot gold standard ($\geq 30\%$ identity, $\geq 50\%$ coverage). Derived metrics: sensitivity 0.53, specificity 0.93, precision 0.72, F1 0.61.

	Gold standard + Gold standard –	
Predicted + (KO assigned)	2,350 (TP)	915 (FP)
Predicted – (no KO)	2,051 (FN)	11,381 (TN)

Table 4. Confusion matrix for *Nannochloropsis gaditana* B-31. KofamScan (predicted) versus the Swiss-Prot gold standard ($\geq 30\%$ identity, $\geq 50\%$ coverage). Derived metrics: sensitivity 0.61, specificity 0.88, precision 0.70, F1 0.65.

	Gold standard + Gold standard –	
Predicted + (KO assigned)	2,156 (TP)	921 (FP)
Predicted – (no KO)	1,353 (FN)	6,499 (TN)

5 Discussion

Two results stand out. First, the gap between the two species in KO yield says more about where the data came from than about the algae themselves: the *D. salina* proteome was pulled from UniProtKB, which barely curates this species, while *N. gaditana* came from a dedicated NCBI project. Second, and more telling, *D. salina* turned out to carry a full set of PUFA desaturases whereas the *N. gaditana* B-31 assembly was missing the $\Delta 4$ - and $\omega 3$ -enzymes that Cunha finds in strain CCMP1894 [6]. B-31 almost certainly still makes these fatty acids; its genome is just less completely annotated. The pipeline flagged this on its own by checking against a published model, which is exactly the kind of thing the Learn phase is meant to catch—and a concrete reason to prefer CCMP1894 for any future work on the EPA pathway.

A word of caution on the numbers. The many pathways that come out as only partially covered are not a sign of missing metabolism: KEGG reference

pathways lump together enzymes from bacteria, plants, fungi and animals, so no single alga was ever going to hit every KO. Coverage is worth reading as a comparison between two species treated identically, not as an absolute score. The same goes for the benchmark—the gold standard is itself just an annotation, so what it really shows is how well KofamScan and the homology search agree. KofamScan is the cautious one, rarely wrong but quieter; the homology search casts a wider net. Between them they cover more ground than either would alone.

6 Conclusion and Future Work

This work set out to compare two microalgae on equal footing, and the DBTL structure made that possible: the same pipeline acquired the data, annotated both proteomes with KofamScan and DIAMOND under Snakemake, benchmarked the result against Swiss-Prot, and cross-checked it against a published model. Along the way it mapped 1,709 shared and 1,730 species-exclusive KOs, characterised the coverage of 397 pathways, and showed that *D. salina* carries a complete PUFA desaturase set while the *N. gaditana* B-31 assembly does not. None of these findings is new biology on its own, but they line up with and reinforce the single-species pictures painted by Shah et al. [4] and Chen et al. [5], and they give independent weight to Cunha’s case for preferring CCMP1894 [6]. What is new is that all of it can be reproduced from raw sequences with a single command.

The natural next step is to rerun the analysis on CCMP1894 and see whether the missing desaturases reappear. Beyond that, the annotation could be deepened with eggNOG-mapper and transcriptomic evidence to chip away at the large fraction of hypothetical proteins, and extended towards genome-scale models with merlin [11] to close the loop from annotation to prediction. Fuller containerisation would round off the FAIR compliance [14] the pipeline already aims for.

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